

timing resistors connected between OSC1 and OSC2 pins of HT12E and HT12D, for a given voltage supply, can be found out from the graphs given in the datasheets of the respective chips. Here we have chosen the resistor values for approximately 3kHz frequency of the encoder (HT12E) and 150 kHz of

the decoder (HT12D) at V_{dd} of 5V.

The HT12D receives data from the HT12E on its D_{in} pin, serially. If the address part of the received data matches the levels on A0 through A7 pins four times in succession, the valid transmission (V_{r}) pin goes high. The data on pins AD8 through AD11 of the HT12E appears on pins D8 through D11 of the HT12D. Thus the device acts as a receiver of the 4-bit data (16 possible codes) with 8-bit addressing (256 possible channels).

Once the frequency of the pair is aligned, then on ground of any data pin on the encoder, LED1 on the decoder should light up. You can also check the data transfer on pins AD8 through AD11, which is available on pins D8 through D11 of the decoder once TE pin is momentarily taken low by making it ground.

RF transmitter and receiver

The RF transmitter and receiver modules from Alps India, Mumbai, have been employed for

PARTS LIST	
Semiconductors:	
IC1	- HT12E Holtek encoder
IC2-IC5	- MCT2E optocoupler
IC6	- HT12D Holtek decoder
IC7	- 74LS138 1-of-8 decoder
IC8	- ULN2803 Darlington array
IC9-IC12	- CD4013 flip-flop
IC13	- CD40106 hex inverter
T1-T9	- BC548 npn transistor
D1-D4	- 1N4148 switching diode
D5-D14	- 1N4007 rectifier diode
TX1	- TX-433 RF transmitter
RX1	- RX-433 RF receiver
LED1	- 5mm LED
Resistors (all 1/4-watt, $\pm 5\%$ carbon):	
R1	- 1-mega-ohm
R2-R5	- 10-kilo-ohm
R6	- 100-kilo-ohm
R7-R10	- 100-ohm
R11	- 47-kilo-ohm
R12	- 3.9-kilo-ohm
R13	- 470-ohm
R14	- 1-kilo-ohm
R15-R22	- 2.2-kilo-ohm
RNW1	- 4.7-kilo-ohm, resistor network
Capacitors:	
C1, C2	- 100 μ F, 16V electrolytic
C3	- 10 μ F, 16V electrolytic
Miscellaneous:	
RL1-RL8	- 6V, 1C/O relay
BATT.1	- 6V battery
BATT.2	- 6V, 4.5Ah battery
SL, S2	- On-off switch
DIP-SW1	- 8-way DIP switch
DIP-SW2	- 8-way DIP switch
	- 25-pin, D-type male connector

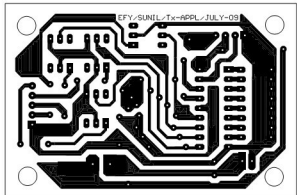


Fig. 5: An actual-size, single-side PCB for the transmitter circuit (Fig. 3)

